Cheatsheet AZ-305 — 12 · Azure Migrate and migration strategies

1. The 5 Rs (strategies)

Strategy	What it is	Typical target	Signal
Rehost (lift and shift)	Move as is, no changes	VMs, AVS	"minimize changes", "quickly"
Refactor (replatform)	Minimal changes for PaaS	App Service, MI, Container Apps	"minimize code changes, use PaaS"
Rearchitect	Redesign (microservices, serverless)	AKS, Functions, Cosmos DB	"modernize", "cloud-native"
Rebuild	Rewrite from scratch	PaaS/serverless	"new application"
Replace	Replace with SaaS	Microsoft 365, Dynamics	"SaaS alternative"

2. Azure Migrate: components

Component	What it does	Exam detail
Azure Migrate project (hub)	Resource in Azure that centralizes discovery, assessment and migration; integrates Microsoft and third-party tools	Created first; discovers nothing by itself
Azure Migrate appliance	VM (OVA for VMware, VHD for Hyper-V, installer for physical) deployed **on-prem**; connects to vCenter / Hyper-V hosts / servers via WinRM-SSH; discovers inventory, **performance** and **dependencies** (agentless)	"what should you configure for the cluster" → appliance
Discovery	Inventory of VMs, SQL Server, web apps (IIS, Tomcat), apps, roles	Continuous while the appliance is active
Assessment	Recommends VM SKU (as-is or **performance-based**), disk, monthly cost, readiness (ready / conditionally / not ready)	"identify resources required", "right-size" → performance-based
Dependency analysis	Agentless (via appliance) or agent-based (Log Analytics + agents); map of what talks to what	"group servers to migrate together", "waves"
Business case	TCO on-prem vs Azure	"justify migration"
Migration and modernization (formerly Server Migration)	Replicates and migrates VMware VMs (agentless or agent-based), Hyper-V, physical, other clouds; **test migration** without affecting production	"test the migration before cutover"
Azure Migrate: Database (DMS)	SQL Server assessment and migration	See section 4
Web app assessment / migration	.NET and Java apps to **App Service** or AKS	"migrate IIS web apps to App Service"
Data Box	Offline migration of massive data	"TB with limited bandwidth"

- Flow: **project → appliance → discovery → assessment (+dependencies) → replicate → test migration → migrate (cutover)**.
- **Azure VMware Solution (AVS):** VMware private cloud **inside Azure** (dedicated hosts); to keep using vSphere/vCenter without converting. Move tool: **HCX**. Choose AVS only if the wording asks to keep VMware.
- **Azure Migrate vs ASR for migrating VMs:** Migrate is the current and recommended tool (uses ASR underneath for VMware agent-based / physical); direct ASR is valid for DR and simple VM migrations.

3. Server migration: which tool

Source	Tool	Mode
VMware vSphere	Azure Migrate (agentless, via appliance)	Snapshot replication; no agent on the VMs
VMware without vCenter access / physical / AWS / GCP	Azure Migrate agent-based (Mobility agent + replication appliance)	Agent on each server
Hyper-V	Azure Migrate (agentless, Hyper-V replication provider on hosts)	No agent on VMs
AWS EC2 / GCP	Azure Migrate agent-based (treated as physical)	Agent
VMware keeping VMware	AVS + HCX	vMotion / bulk migration
DR only or very simple migration	Azure Site Recovery	Continuous replication

4. Database migration

Source → Target	Tool	Downtime	Notes
Assess SQL Server beforehand	**Data Migration Assistant (DMA)** or Azure Migrate SQL assessment	—	Compatibility, deprecated features, SKU recommendation
SQL Server → Azure SQL Database	**BACPAC** (SSMS/sqlpackage) · DMS · **Azure Data Studio + SQL Migration extension**	BACPAC high; DMS/ADS online low	Single DB does not support cross-database
SQL Server → Managed Instance	**Azure Data Studio + SQL Migration ext.** (online/offline) · **DMS** (online/offline, many DBs) · **Log Replay Service** (native backups) · **Managed Instance link** (distributed AG with MI, near-zero downtime)	Minimal with online	50 DBs offline with minimal effort → DMS

Source → Target	Tool	Downtime	Notes
SQL Server → SQL on VM	**Distributed AG** (from an existing AG, cross-version, Windows→Linux) · Log shipping · Backup/restore · ASR (whole VM)	Distributed AG minimal	See SQL VM cheatsheet
Oracle / MySQL / DB2 / SAP ASE / Access → SQL	**SQL Server Migration Assistant (SSMA)**	High	Converts schema and code
MySQL / PostgreSQL → Azure Database Flexible Server	**DMS** (online with replication) · dump/restore · **read replica** from on-prem does not apply	Minimal with DMS online	
MongoDB → Cosmos DB for MongoDB	DMS · Cosmos DB migration tool · mongodump/mongorestore	Variable	
Cassandra → Cosmos DB for Cassandra	Dual-write proxy · Spark		
Massive data (TB) to Storage/ADLS	AzCopy · Data Factory · **Data Box** / **Import-Export**		See cheatsheet 03
SSIS packages	**Azure-SSIS Integration Runtime** in ADF; SSISDB in Azure SQL Database (cost) or MI (compatibility)		No changes to packages
Continuous on-prem ↔ Azure access (not migration)	**Self-hosted Integration Runtime** (formerly Data Management Gateway)		ADF/Synapse with on-prem sources

- **The words "minimize downtime"** → rule out BACPAC, backup/restore, copying VHDs; look for continuous replication (DMS online, ADS online, Distributed AG, MI link, ASR).
- **The word "offline"** → DMS offline or backup/restore; the criterion becomes effort and number of DBs.

5. Application migration

Scenario	Target	Tool / pattern
IIS .NET web app with no OS access required	App Service (Windows)	Azure Migrate web app migration / App Service Migration Assistant
App with COM components, OS dependencies	VM / VMSS	Rehost
App with LDAP against AD DS, no VPN allowed	VM + **Entra Domain Services**	Identity refactor
On-prem app with IWA/Kerberos for remote users	Stays on-prem + **Application Proxy (KCD)**	No migration
Microservices in containers	AKS / Container Apps	Rearchitect
Functions Consumption with queue trigger → AKS	AKS + **KEDA + HPA**	Replatform
Java Spring apps	Azure Spring Apps / Container Apps	Refactor
In-memory session state → scalable	App Service + **Azure Cache for Redis**	Refactor

6. File and File Server migration

Scenario	Solution
On-prem file server, keep local + cloud access	**Azure File Sync** (cloud tiering)
Migrate shares to Azure Files	Azure File Sync · Storage Migration Service (Windows Admin Center) · Robocopy / AzCopy
Massive data, little bandwidth	**Data Box** (Microsoft ships it) / **Import-Export** (your disks)
Continuous transfer from on-prem	Data Box Gateway
Rarely accessed files, minimum cost	Blob **Archive** (24 h rehydration acceptable)

7. Decision tree

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What am I migrating?
├─ VMware / Hyper-V / physical VMs ....... Azure Migrate (appliance → assessment → migrate); AVS if they want to stay on VMware
├─ Only assess SQL Server ................ DMA / Azure Migrate SQL assessment
├─ SQL → Single DB ....................... BACPAC (downtime OK) / ADS-DMS online (min downtime)
├─ SQL → Managed Instance ................ ADS + SQL Migration ext. (online) / DMS (many DBs, offline) / MI link
├─ SQL → SQL on VM from an AG ............ Distributed AG
├─ Oracle/MySQL/DB2 → SQL ................ SSMA
├─ MySQL/PostgreSQL → Flexible Server .... DMS online
├─ SSIS ................................ Azure-SSIS IR + SSISDB in Azure SQL DB
├─ IIS web app ......................... App Service (Migrate web app tool)
├─ Files, TB, little bandwidth ........... Data Box / Import-Export
└─ Continuous access, not migration ...... Self-hosted IR / On-premises Data Gateway
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8. Signals → answer

Wording	Answer
"VMware cluster, identify resources required, minimize effort"	Azure Migrate appliance (project as hub)
"keep using vSphere / VMware tools"	Azure VMware Solution

Wording	Answer
"group servers by dependencies"	Dependency analysis
"right-size based on actual usage"	Performance-based assessment
"test before cutover"	Test migration
"50 databases, MI, offline, min effort"	Database Migration Service
"SQL 2008 → MI, minimize downtime"	Azure Data Studio (SQL Migration extension)
"existing AG → SQL 2019 on Linux VM, minimize downtime"	Distributed availability group
"Oracle to Azure SQL"	SSMA
"SSIS packages, no hybrid connectivity, min cost"	Azure-SSIS IR + SSISDB in Azure SQL Database
"5 TB rarely accessed files"	Blob Archive
"10 TB, limited bandwidth"	Data Box
"file server with local cache"	Azure File Sync

9. Mistakes to avoid

1. Azure Migrate as "the tool that migrates databases" → it is the hub; DMS/ADS do the migrating.
2. DMA to migrate to MI → DMA only assesses (and migrates to Single DB).
3. SSMA for SQL Server → SQL → SSMA is for **other engines**.
4. AVS when native Azure VMs are required.
5. BACPAC with "minimize downtime".
6. Copying VHDs when ASR / Azure Migrate exist.